

Ivan Anokhin

PhD Student

Montreal

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Research interests Scalable reinforcement learning systems; asynchronous neural network computation; RL for LLM agents; mid- and long-term memory mechanisms.

Education

08.2023 – 09.2027 PhD in Computer Science, Mila, University of Montreal, Advisor: Irina Rish, Topic: Scalable Reinforcement Learning Systems.

2017 – 2019 Master in Data Science, Skolkovo Institute of Science and Technology, Advisor: Dmitry Yarotsky, Thesis: Loss Surface of a Deep Neural Network.

2012 – 2016 BSc in Applied Mathematics and Computer Science, St. Petersburg State University.

Selected Publications

Under review **Learning From the Past with Cascading Eligibility Traces.**
Tokiniaina Raharison Ralambomihanta*, **Ivan Anokhin***, Roman Pogodin*, Samira Ebrahimi Kahou, Jonathan Cornford, Blake Aaron Richards
Paper, Code

ICML2025 workshop **AIF-GEN: Open-Source Platform and Synthetic Dataset Suite for Lifelong Reinforcement Learning on Large Language Models.**
Jacob Chmura*, Shahrad Mohammadzadeh*, **Ivan Anokhin**, Jacob-Junqi Tian, Mandana Samiei, Taz Scott-Talib, Irina Rish, Doina Precup, Reihaneh Rabbany, Nishanth Anand
Paper, Code

ICLR 2025 **Handling Delay in Real-Time Reinforcement Learning.**
Ivan Anokhin, Rishav Rishav, Matthew Riemer, Stephen Chung, Irina Rish, Samira Ebrahimi Kahou
Paper, Code, Blog post

NeurIPS 2023 **Thinker: Learning to Plan and Act.**
Stephen Chung, **Ivan Anokhin**, David Krueger.
Project Page

Experience

08.2023–now PhD, Mila.

Async forward pass: Parallel layer-wise inference for streaming/on-device RL; proposed a method to handle the induced delay; led experiments, theory, and writing; accepted to ICLR 2025.

Async backward pass: Bio-inspired credit assignment with delayed feedback (parallel layer-wise backward pass); originated the idea and led RL experiments; currently under submission.

Continual RLHF: Contributed to continual-RLHF benchmark; benchmarked DPO/PPO and implemented evaluation metrics; published in ICML 2025 workshop.

LLM reasoning distillation: Exploring a teacher–student framework with off-policy RL training (TOPR); work in progress.

- 02.2023–09.2023** Research Assistant (remote), **Cambridge University**.
Model-based RL: Worked on model-based offline RL, contributed to planning project, published in NeurIPS2023.
- 06.2021–11.2022** Research Scientist, **Yandex**.
Self-supervised learning: Developed a framework to learn image representations object-wise for unsupervised object detection. Led a team of two students. Worked on an SSL method to learn representations from long videos (e.g. EPIC Kitchen).
- 03.2020–01.2022** Junior Research Scientist, **Skoltech**.
Loss landscape and ensembles: Investigated properties of the ensemble methods with reduced memory consumption at inference and high accuracy (e.g. BatchEnsemble). Published in AISTATS2022. Investigated loss surface of neural networks. Published in ICML2020.
- 04.2019–06.2021** Deep Learning Engineer, **Samsung AI Center**.
GANs: Developed a generator architecture for GANs without any pixel-interaction after style (noise) conditioning. Published in CVPR2021. Developed generative image-to-image model with style (day-time of an image) and content (landscape) disentanglement. Published in CVPR2020.
Robot navigation: Worked on robot navigation in an indoor environment. Added physics into the environment simulator; debugged a robot in a real environment.
- 12.2016–08.2018** Lead Analyst (ML), **Tinkoff Bank**.
Developed, implemented and tuned Slot-filling and Information Retrieval (IR) transformer-based neural network architectures for the chatbot. Slot-filling and IR models were deployed.

Technical skills

Proficient in: Python, Pytorch, git, Latex
Familiar with: Ray, vLLM, Deepspeed, Jax, Keras, Tensorflow, SQL, Java, Matlab, C++

Service and Teaching

Reviewer ICML21-22, ICLR22-23, NEURIPS22-25
Lecturer Invited lecturer on deep learning, GoTo summer camp for high-schoolers, 2020 & 2021 multi-day sessions.